Lab05

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Background

There are lot's of ways to make plots in R. These include so-called "base R" (like the `plot()` function) and add on packages like **ggplot2**

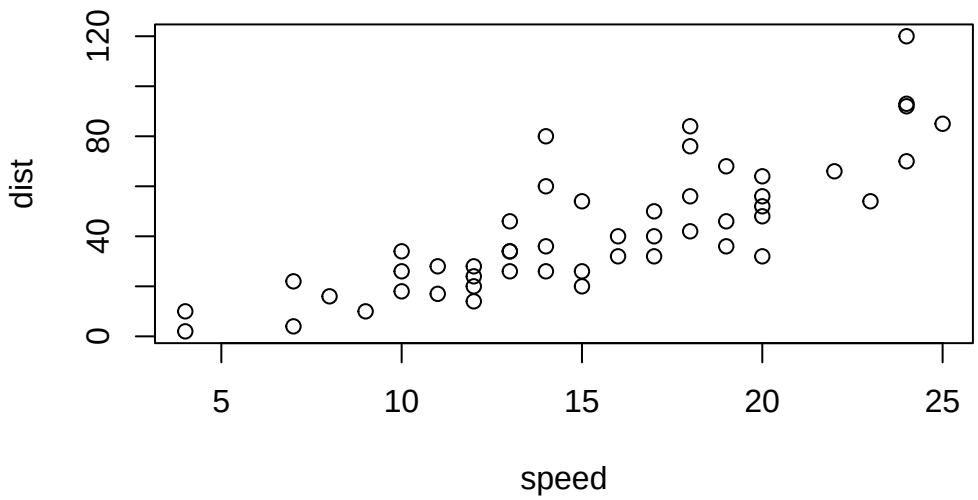
Let's make the same plot with these two graphics systems. We can use the inbuilt `cars` data set:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

With "base R" we can make graphs using the `plot()` function

```
plot(cars)
```



Now using ggplot, first install the package using `install.packages(ggplot2)`

> **N.B.** do not place install.packages in a code chunk otherwise it repeatedly installs every time the document is rendered.

after installation of the library, the library needs to be loaded with a call to the function `library()`

```
#install.packages("ggplot2")  ggplot installation code
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

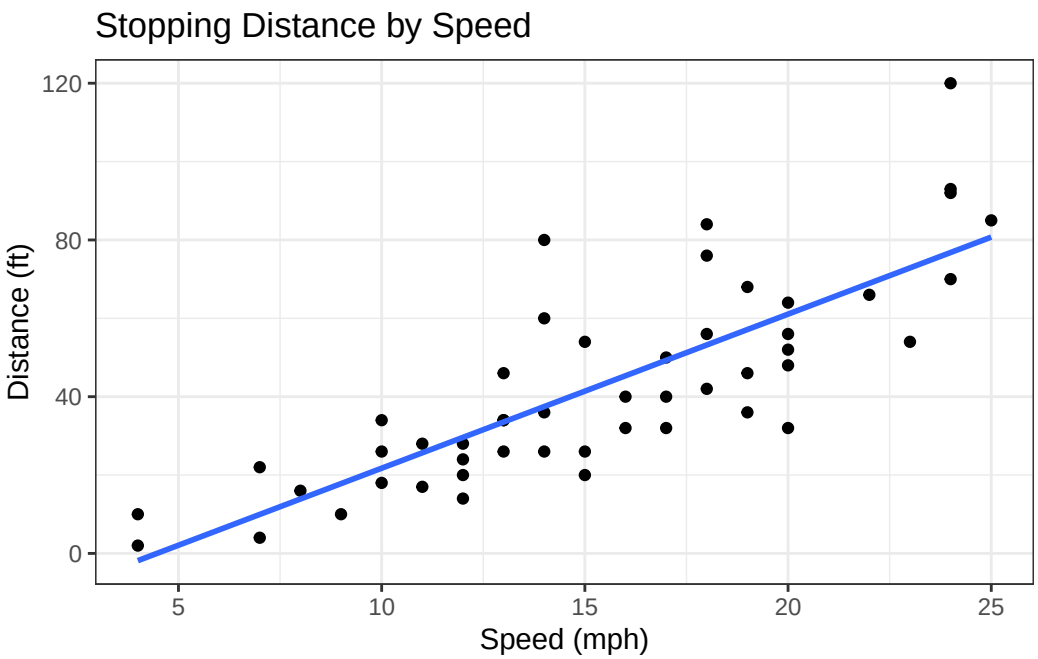
ggplot objects need at least 3 things:

1. the `data` to use to be graphed
2. the `aes` for aesthetics that map the data of the graph
3. the `geom` for the plot type

```
  ggplot(data = cars) +
  aes(x = speed, y = dist) +
  geom_point() +
  geom_smooth(method = "lm", se = F) +
```

```
  labs(x = "Speed (mph)", y = "Distance (ft)", title = "Stopping Distance by Speed") +
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Gene Expression Plot

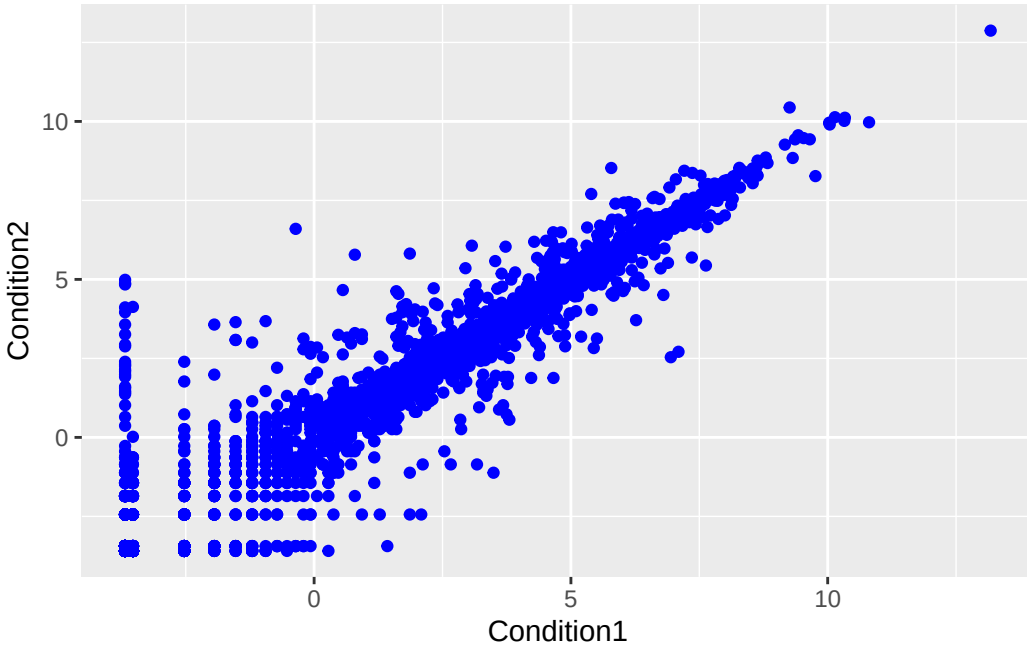
Using data on gene expression with and without GLP-1 inhibitors we can plot a graph to highlight significant genes that are up or down regulated.

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

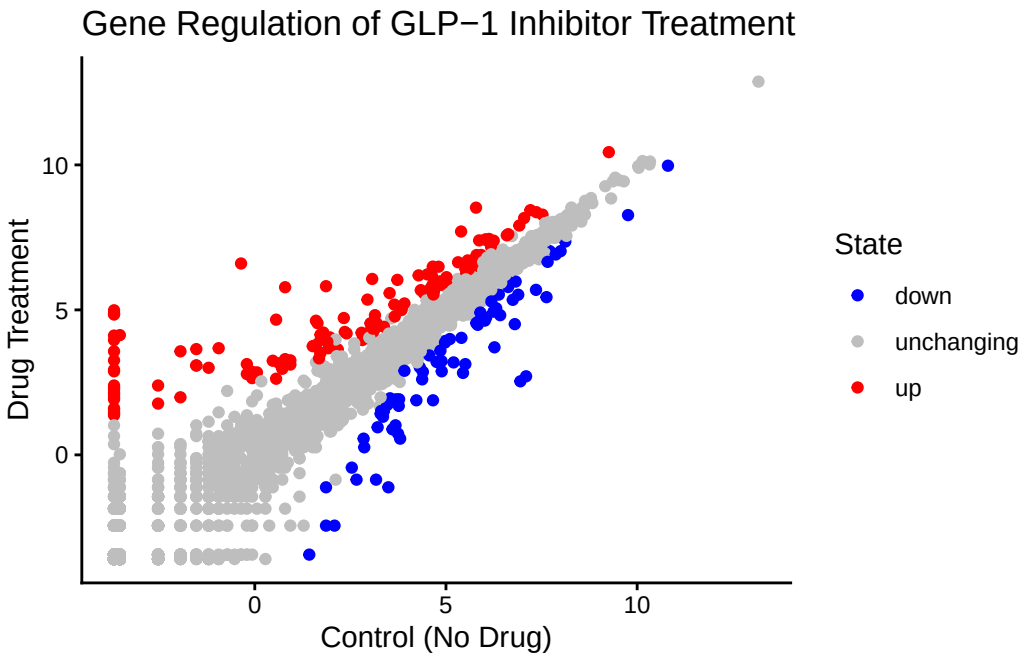
version 1 plot - simple dot plot comparing gene expression with and without treatment.

```
ggplot(genes) +
  aes(x = Condition1, y = Condition2) +
  geom_point(col = "blue")
```



Using the `State` value for each gene, we can color the graph based on the state within `aes()` and specify the colors using `scale_color_manual()`

```
ggplot(genes) +
  aes(x = Condition1, y = Condition2, col = State) +
  geom_point() +
  scale_color_manual(values = c("blue", "gray", "red")) +
  labs(x = "Control (No Drug)",
       y = "Drug Treatment",
       title = "Gene Regulation of GLP-1 Inhibitor Treatment") +
  theme_classic()
```



Further Exploration with gapminder

Using the `gapminder` data set we look to explore further capabilities of ggplot2

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv
gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many rows (total entries) does the data set have

```
nrow(gapminder)
```

```
[1] 1704
```

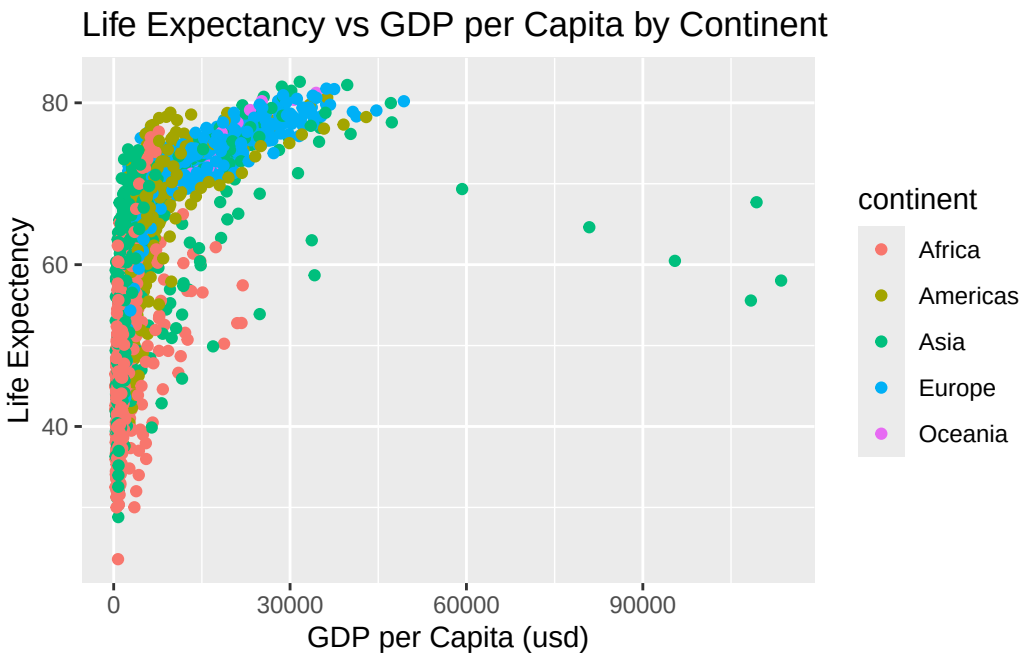
how many different continents are in the data set

```
unique(gapminder$continent)
```

```
[1] "Asia"     "Europe"   "Africa"   "Americas" "Oceania"
```

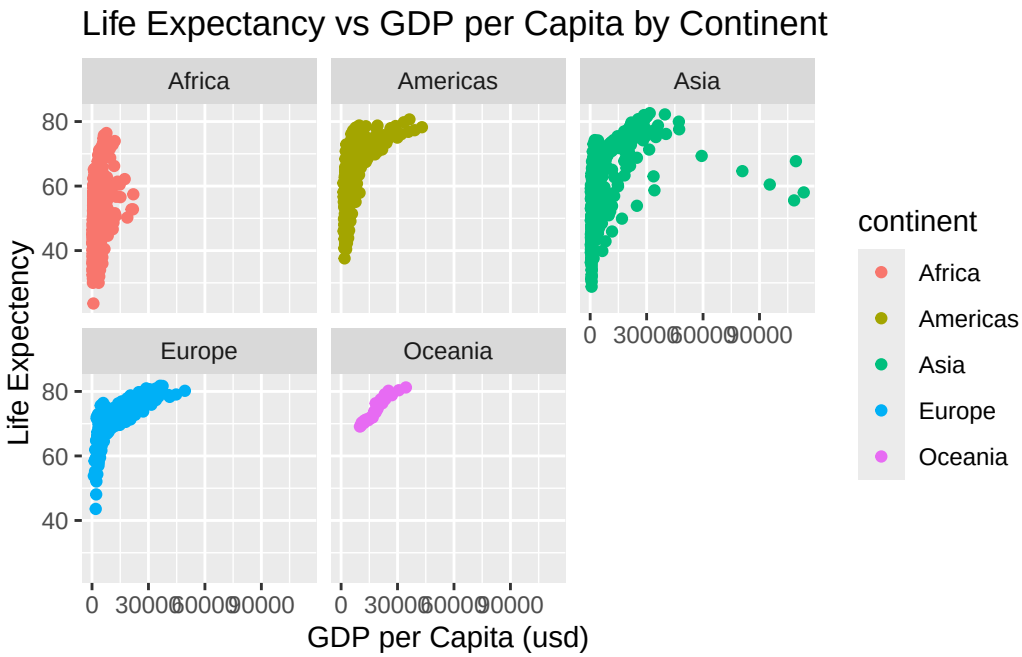
Version 1 plot: `gdpPercap` vs `lifeExp`

```
ggplot(gapminder) +
  aes(x = gdpPercap, y = lifeExp, col = continent) +
  geom_point() +
  labs(x = "GDP per Capita (usd)", y = "Life Expectency", title = "Life Expectancy vs GDP per
```



Creating separate plots for each continent using the "faceting" feature in ggplot:

```
ggplot(gapminder) +
  aes(x = gdpPercap, y = lifeExp, col = continent) +
  geom_point() +
  facet_wrap(~continent) +
  labs(x = "GDP per Capita (usd)", y = "Life Expectency", title = "Life Expectancy vs GDP per
```



Using the dplyr library

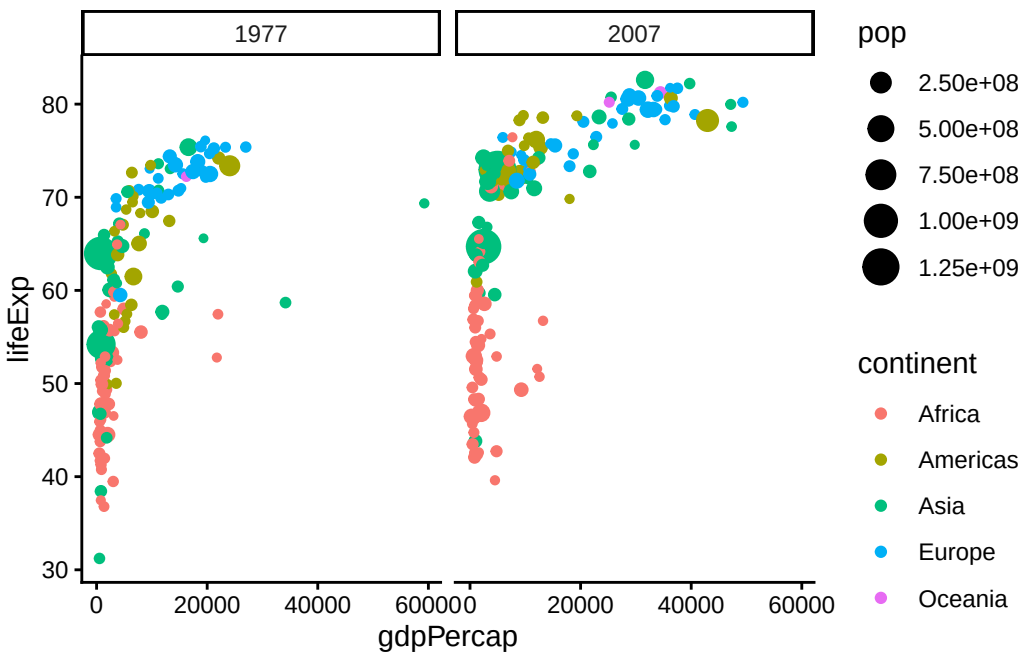
Adding another package `dplyr` that allows for the usage of the `filter()` function to select data points from the data set

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
filter(gapminder, year == 2007, country == "United States")
```

```
        country continent year lifeExp       pop gdpPercap
1 United States  Americas 2007  78.242 301139947  42951.65
```

```r
input <- filter(gapminder, year == 2007 | year == 1977)
```

```r
ggplot(input) +
  aes(x = gdpPercap, y = lifeExp, col = continent, size = pop) +
  geom_point() +
  facet_wrap(~year) +
  theme_classic()
```